Hand Gesture Controlled Robot 🤖✋

A wireless robot car controlled by hand gestures using an ESP32, MPU6050 accelerometer, and Bluetooth communication.

🌟 Features
* **Gesture Control:** Tilt forward/back to move, tilt left/right to turn.
* **Master-Slave Bluetooth:** ESP32 (Master) talks to HC-05 (Slave).
* **Driver Logic:** L298N motor driver with custom steering logic.

🔌 Circuit Connections

1. Receiver Module (The Car)
Microcontroller: Arduino Uno

A. Motor Driver (L298N)
L298N Pin	Arduino Pin / Connection
IN1	Pin 8
IN2	Pin 9
IN3	Pin 10
IN4	Pin 11
ENA / ENB	Jumper Caps ON (Full Speed)
12V	Battery (+)
GND	Battery (-) **AND** Arduino GND

B. Bluetooth Module (HC-05)
HC-05 Pin	Arduino Pin
VCC	5V
GND	GND
TX	Pin 0 (RX)
RX	Pin 1 (TX)

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2. Transmitter Module (The Glove)
Microcontroller: ESP32 Dev Module

MPU6050 Sensor
MPU6050 Pin	ESP32 Pin
VCC	3.3V
GND	GND

| **SCL** | GPIO 22 |
| **SDA** | GPIO 21 |

🚀 How to Run
1. **Upload Receiver Code:** * Open `Receiver_Code.ino`.
   * **Disconnect HC-05 wires (Pin 0 & 1) before uploading.**
   * Upload to Arduino Uno. Reconnect wires.
2. **Upload Transmitter Code:**
   * Open `Transmitter_Code.ino`.
   * Upload to ESP32.
3. **Power On:**
   * Turn on the Car first, then the Glove.
   * They will auto-pair (Password: `1234` is hardcoded).
